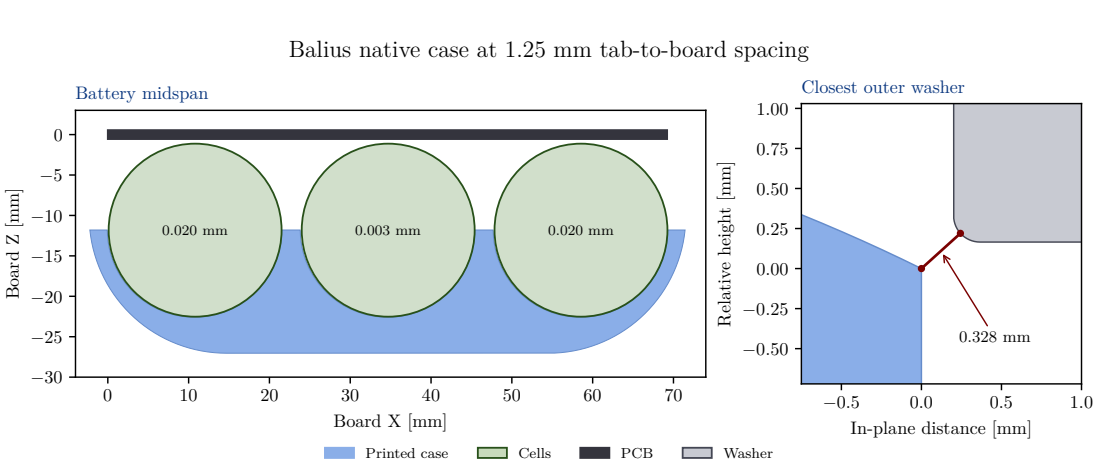
The three-pocket Balius case fits at 1.25 mm stand-off



The middle-cell clearance is 0.003 mm and the closest washer clears by 0.328 mm. The rigid left cell reaches its pocket after **0.020 mm toward the middle cell** or **0.165 mm away from it**.

The shell adds 73.7 g to the 251.8 g bare assembly

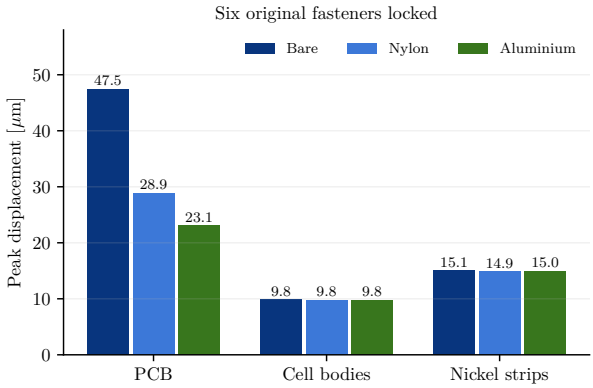
Snug geometry
Three cells, four 6.35 mm thick tabs and four M4 case screws. The nominal model uses printed PLA, an air gap, and either nylon or aluminium screws.

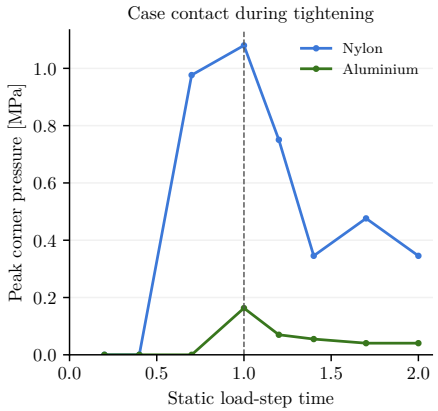
The shell mass assumes dense PLA. Screws and spacer mass are excluded from the estimate.

No-case reference
First prestressed mode 485.96 Hz. Peak board-normal motion under base Z was 52.67 μm RMS. At 0.50 mm lateral travel the bare fixture required 8.54 N in positive X and 8.51 N in negative X.

The rerun compares the same loads and movie subjects. The earlier flush and 3 mm results are preserved in Appendix A of the report.

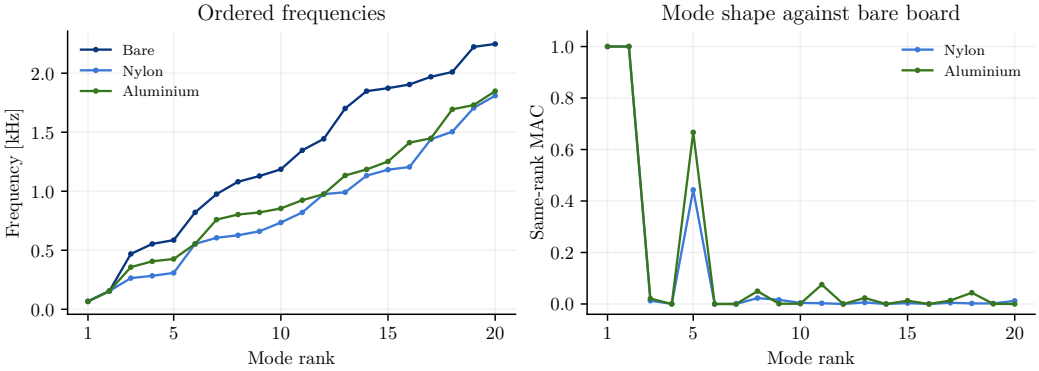
Both snug mounts reduce PCB movement during tightening





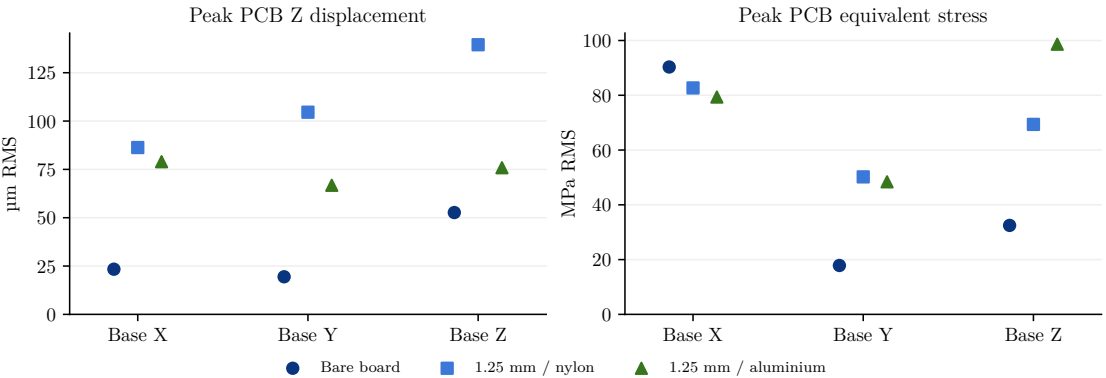
At bolt lock, peak PCB displacement fell from **47.46 μm bare** to **28.91 with nylon screws** and **23.09 with aluminium screws**. All three nylon meshes put the peak at the same corner, with a spread of 0.055 μm. Only the middle cell touched the case.

The first unpreloaded mode stays at 66.61 Hz



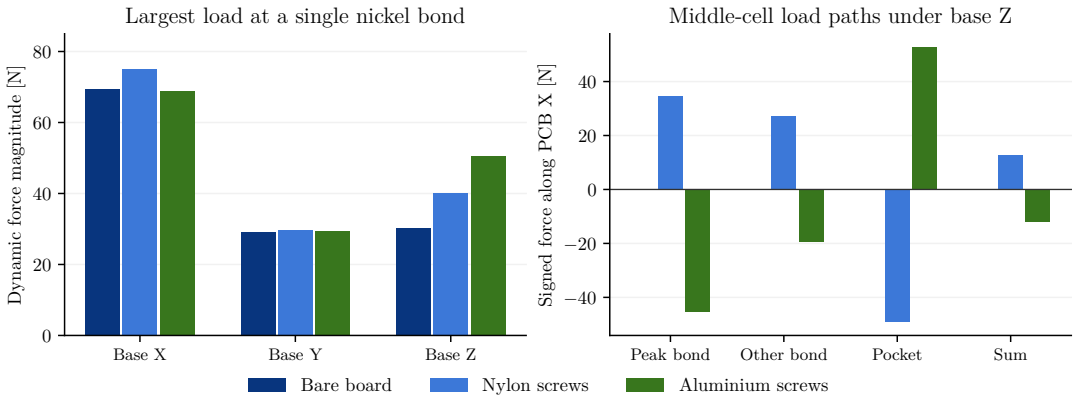
The first two mode shapes closely match the bare board. At mode 20 the shapes differ, so comparing frequencies by rank alone would give a misleading shift. These are the refined default-contact controls before bolt tightening. Paired movies.

Aluminium cuts board travel relative to nylon but raises base-Z stress



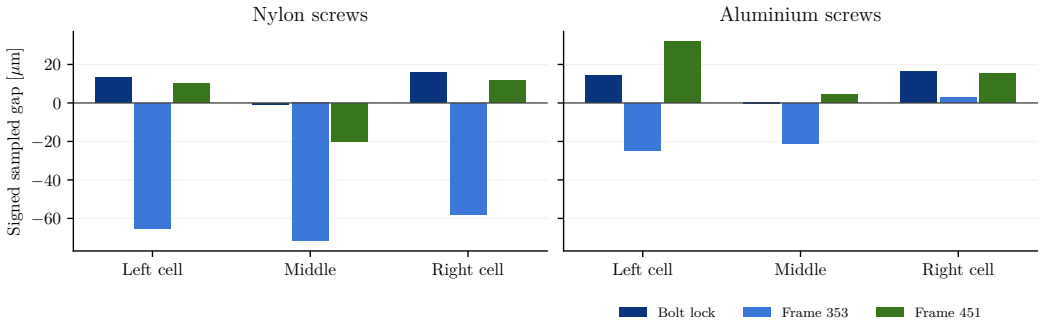
Under base Z, peak PCB stress reached **32.5 MPa RMS bare**, **69.4 with nylon** and **98.7 with aluminium**. Under base Y both mounts raised stress over nearly the entire board. These native spectra hold the installed contact state fixed.

Board-normal vibration loads the middle cell's nickel bond



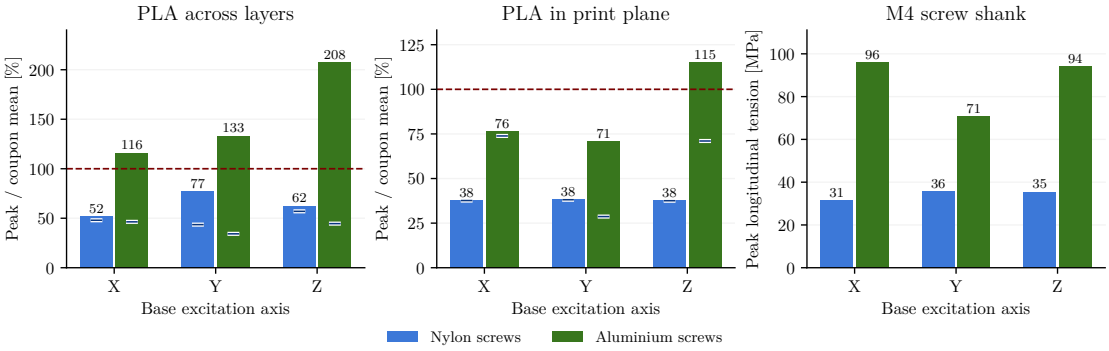
The largest base-Z bond load rose from **30.2 N bare** to **40.1 N with nylon screws** and **50.5 N with aluminium**. The governing bond moved from the right cell to the middle cell. The bars use independent integration on native modes with the installed contacts held fixed.

The outer cells can reach the case



At bolt lock, the two finer nylon meshes carried **0.57–0.59 N** at the middle pocket. Both meshes require the installed pressure to redistribute within about **13 ms** under base Z. Under base Y, the bounds are 25.6 and 21.0 ms. The time of cell lift-off remains unresolved.

The printed tab exceeds its coupon reference with aluminium screws



On the nominal mesh the tab reached **77% of the through-layer coupon mean with nylon** and **208% with aluminium**. Refining the nylon case shifted its maximum to **71.5% under base X**. The aluminium mesh check is queued. Horizontal marks exclude the 2 mm bearing band.